Initial KE_2 = 4.0 / 2 = 2.0 MeV

30 Ans: C

Excluding the background count-rate of 16 min^{-1}, the initial total count-rate of the α-particles and β-particles is 336 min^{-1} (= 352 – 16).

The sheet of paper prevents the α-particles from reaching the detector. Hence the initial count rate due to the α-particles is 96 min^{-1} (= 352 – 256). The initial count rate due to the β-particles is 240 min^{-1} (= 336 – 96).

At t = 12 days,

count-rate of α-particles = $\left(\frac{1}{2}\right)^{\frac{12}{4}}(96)$

count-rate of β-particles = $\left(\frac{1}{2}\right)^{\frac{12}{3}}(240)$

total count-rate (including background) = $\left(\frac{1}{2}\right)^{\frac{12}{4}}(96) + \left(\frac{1}{2}\right)^{\frac{12}{3}}(240) + 16 = 43\ \text{min}^{-1}$